BHANU PRAKASH REDDY VELLALA LINGA

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PROFESSIONAL SUMMARY

Generative AI Engineer and Full Stack AI Developer with over 2 years of hands-on experience designing and delivering production-grade enterprise AI applications using LLMs, Agentic AI, Retrieval-Augmented Generation (RAG), and multi-agent systems. Proficient in Python, FastAPI, Azure OpenAI, Semantic Kernel, and LangChain, with hands-on expertise in vector embeddings, hybrid search, prompt engineering, LLM fine-tuning, and AI observability. Delivered scalable AI solutions across content, finance, manufacturing, and medical domains processing 1M+ digital assets and enabling natural language data access for 100+ business users.

TECHNICAL SKILLS

Generative AI & LLMs: Azure OpenAI, OpenAI GPT-4, AWS Bedrock, Gemini, RAG, Agentic RAG, Multi-Agent Systems, LLM Fine-tuning, Prompt Engineering

AI Frameworks: Semantic Kernel, LangChain, Google ADK, PromptFlow

Programming & APIs: Python, FastAPI, REST APIs, Async Processing, Pydantic, ReactJS, TypeScript

Databases & Vector Stores: SQL Server, PostgreSQL, MongoDB, Azure Cosmos DB, Pinecone, ChromaDB, Vector Embeddings

Search & Cloud: Azure AI Search, AWS OpenSearch, Elasticsearch, Azure Functions, Logic Apps, Azure AI Foundry, Docker, CI/CD

Observability & Tools: PromptFlow, Azure Monitor, LLM Evaluation, Token Optimization, Git, GitHub Copilot, VS Code

PROFESSIONAL EXPERIENCE

Mouri Tech Ltd — Hyderabad, India

Generative AI Engineer – Agentic & Applied AI (Full Stack) — *May 2024 – Present*

Client: *Stanley Black & Decker, USA*

- Architected and delivered 6 production-grade AI applications using LLMs, Agentic RAG, multi-agent systems, and Azure OpenAI across content, finance, manufacturing, and medical domains.
- Engineered scalable RAG pipelines with hybrid search and vector embeddings, improving document retrieval accuracy by 42% and reducing LLM inference latency by 35%.
- Optimized token usage, async processing, and caching strategies — cutting compute costs by 28% across enterprise deployments.
- Integrated PromptFlow and Azure Monitor for AI observability, LLM evaluation, and real-time workflow analytics; collaborated with cross-functional teams to deliver solutions on schedule.

KEY PROJECTS

Multi-Legal Alt Text Generation Platform | *FastAPI, ReactJS, Azure OpenAI, Cosmos DB, Docker, Bynder DAM*

- Built an end-to-end AI platform integrating Azure OpenAI with Bynder DAM to auto-generate legally compliant, multilingual alt-text for 1M+ enterprise images across 10+ languages — saving ~2,500 hrs/year of manual tagging effort.
- Designed a scalable batch-processing pipeline using FastAPI with async job queuing, enabling high-throughput image processing and seamless Azure OpenAI integration for automated multilingual content generation at enterprise scale.
- Developed a React + TypeScript dashboard for content teams to monitor job queues, review AI-generated alt-text, and manage approval workflows before publishing back to Bynder DAM — reducing content review cycle time by 60%.

Finance AI Agent – Natural Language Data Access | *Semantic Kernel, FastAPI, Azure OpenAI, SQL Server, PostgreSQL, Cosmos DB*

- Developed a conversational LLM agent using Semantic Kernel and Azure OpenAI with Text-to-SQL generation, enabling 100+ non-technical finance users to query revenue, invoices, and payment data in plain English — eliminating SQL dependency entirely.
- Implemented multi-turn context memory and dynamic SQL query construction across SQL Server and PostgreSQL, supporting real-time financial analytics, invoice tracking, payment monitoring, and executive dashboard reporting.
- Reduced financial reporting turnaround from several hours to under 2 minutes by designing intelligent conversational workflows with query refinement, drill-down capabilities, and AI-generated data summaries for business stakeholders.

Manufacturing – AI Sales Cross Reference Copilot | *Semantic Kernel, Azure OpenAI, Azure AI Search, Snowflake, React*

- Developed a Generative AI-powered sales copilot using Semantic Kernel and Azure OpenAI for intelligent product comparison, cross-referencing across thousands of SKUs, and AI-driven pricing recommendations — reducing product lookup time by 50% for sales teams.
- Integrated Azure AI Search semantic retrieval with Snowflake real-time data pipelines to power intelligent product matching, substitute identification, and contextual pricing optimization across a large enterprise product catalog.
- Built an interactive React UI dashboard for sales representatives to perform natural language product searches, view AI-driven cross-reference suggestions, and access pricing decision support — improving sales team efficiency and accelerating deal closure time.

Medical – Handwriting Extraction Automation | *Azure OpenAI, Logic Apps, Function App, Blob Storage*

- Automated extraction and digitization of handwritten clinical data from emails and scanned documents using Azure OpenAI multimodal capabilities — cutting manual data entry effort by 80%.
- Built an event-driven pipeline with Azure Logic Apps and Function Apps for hands-free document ingestion, AI processing, and structured EHR-ready output storage in Azure Blob Storage.

Smart Gen AI Studio – Enterprise RAG Platform | *Azure OpenAI, AWS Bedrock, Azure AI Search, AWS OpenSearch, FastAPI*

- Architected a multi-cloud RAG platform connecting SharePoint and enterprise data sources to LLM-powered hybrid semantic search, boosting knowledge retrieval precision by 38%.
- Integrated Azure OpenAI and AWS Bedrock as interchangeable LLM backends for multi-cloud flexibility and cost-optimized inference routing based on query complexity.

Marketing – AI Document Summarization | *Azure AI Foundry, FastAPI, Azure Function App*

- Built an async document summarization microservice using Azure AI Foundry and advanced prompt engineering, reducing business report review time by 70% and delivering instant executive summaries on document upload via event-driven triggers.

EDUCATION

B.Tech – Electrical & Electronics Engineering 2019 – 2023

Annamacharya Institute of Technology & Sciences, YSR Kadapa, India